

Notes: Mayer, Piskorski & Tchisty (JFE, 2013)
The inefficiency of refinancing: Why prepayment penalties are good for risky borrowers

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1 Big picture

- **Question.** Are prepayment penalties in risky mortgage markets necessarily predatory, or can they improve welfare for risky borrowers?
- **Main answer.** In a competitive dynamic lending model with costly default, refinancing penalties can improve borrower welfare, lower mortgage rates, reduce defaults, and expand credit availability for risky borrowers.
- **Key mechanism.** Free refinancing lets borrowers whose creditworthiness improves leave the original pool. Lenders anticipate this adverse selection and charge higher initial rates. Higher rates hurt borrowers who later receive bad shocks and can exclude high-risk borrowers from credit markets.
- **Role of the penalty.** A prepayment penalty functions as a commitment device: it limits opportunistic refinancing and preserves risk sharing within the original borrower risk class.
- **Riskier borrower result.** The gains from commitment are largest for low-credit-quality borrowers because they benefit most from future positive credit shocks.
- **Mobility trade-off.** Prepayment penalties can inefficiently restrict moving. Therefore, the optimal penalty should be contingent: high for refinancing, but low or zero for house sales.
- **Empirical strategy.** The paper studies fixed-rate mortgages originated in June 2003, when conforming mortgage rates were near the lowest level in 1988–2008.
- **Main evidence.** Positive house price and credit-score shocks predict mortgage prepayment; the effect is much larger for subprime loans; prepayment penalties are concentrated among riskier borrowers; and penalty loans carry lower coupon rates for subprime borrowers.

2 Incremental contribution of the paper

- **Reframes a controversial contract term.** The paper challenges the standard view that subprime prepayment penalties are mainly evidence of exploitation.
- **Adds credit-quality refinancing to the prepayment literature.** Much mortgage prepayment work focuses on interest-rate declines; this paper focuses on refinancing after borrower-specific credit improvement.
- **Connects mortgage design to reclassification risk.** If good risks can exit costlessly, bad risks are left with higher prices. Prepayment penalties restore some long-term risk sharing.
- **Explains concentration among risky borrowers.** The model predicts that penalties should be used most by the borrowers who look riskiest ex ante.
- **Separates refinancing from mobility.** Penalties on refinancing can be efficient while penalties on house sales can be costly, motivating soft or contingent penalties.
- **Links theory to targeted evidence.** The empirical design uses June 2003 FRMs and tests the model predictions with LoanPerformance and BlackBox-Equifax data.

3 Data and measurement

The main data are loan-level LoanPerformance data on securitized mortgages. The baseline sample focuses on owner-occupied fixed-rate mortgages originated in June 2003 with known prepayment penalty status and MSA house price data. Prime loans are identified by prime mortgage-backed security pools. Subprime loans are from subprime pools and have origination FICO below 620.

Interpretation. The data are chosen to match the model: fixed-rate loans, a low-interest-rate origination month, and borrowers whose prepayment behavior can be linked to changes in creditworthiness rather than simple rate refinancing.

3.1 Key variables

- **Pays Off in X Quarters:** equals one if the loan is voluntarily prepaid in the first eight or sixteen quarters after origination.
- **Average Quarterly HPI Growth:** MSA-level house price growth, used as a proxy for positive borrower wealth/collateral shocks.
- **Prepayment Penalty:** equals one if the loan contains a prepayment penalty clause.
- **Coupon Rate:** monthly mortgage interest rate charged to the borrower.
- **FICO/Vantage:** borrower credit quality at origination and over time.
- **CLTV:** combined loan-to-value; junior liens and high CLTV loans are especially risky.

4 Model and empirical specifications

4.1 Borrower and lender primitives

$$\text{Lender value} = E \left[\int_0^\infty e^{-rt} df_t \right], \quad \text{Borrower value} = E \left[\int_0^\infty e^{-\gamma t} dC_t \right], \quad \gamma > r.$$

Interpretation. The borrower is more impatient than the lender. This creates value from financing homeownership but also makes costly default relevant.

4.2 Borrower income/wealth process

$$dY_t = \mu dt + \sigma dZ_t.$$

Interpretation. The borrower's creditworthiness evolves stochastically. Positive shocks improve refinancing prospects; sufficiently negative shocks trigger costly default.

4.3 No-refinancing versus freely prepayable mortgage

A sufficiently high prepayment penalty implements the no-refinancing contract. Free refinancing allows good-shock borrowers to leave the pool, so the lender prices the original loan anticipating that bad-shock borrowers remain.

Free refinancing $\Rightarrow$ higher initial coupon $\Rightarrow$ more defaults and possible exclusion of risky borrowers.

Interpretation. The penalty is a commitment mechanism that permits lower initial rates and better pooling of credit risk.

4.4 Empirical payoff/prepayment regressions

$$\Pr(\text{PaysOff}_i = 1) = F(\alpha + \beta \text{HPIgrowth}_i + \gamma' X_i + \varepsilon_i), \quad \beta_{\text{subprime}} > \beta_{\text{prime}} > 0.$$

Interpretation. Positive collateral or wealth shocks should induce refinancing, especially for risky borrowers.

4.5 Coupon-rate regression

$$\text{Coupon}_i = \alpha + \beta \text{PrepaymentPenalty}_i + \gamma' X_i + \text{MSA FE} + \varepsilon_i, \quad \beta < 0.$$

Interpretation. Penalty loans should carry lower rates, with a larger discount for riskier junior liens.

5 Identification and empirical design

The empirical design narrows the setting to June 2003 fixed-rate mortgages so that observed prepayments are less likely to be driven by falling aggregate mortgage rates. House price appreciation and Vantage-score changes proxy for borrower credit improvement. Senior versus junior lien status provides a risk gradient for testing whether rate discounts from penalties are larger for riskier loans.

6 Tables and figures

6.1 Figure 1: Mortgage premiums and borrower net utility gain

Read: The left panel shows that free refinancing raises required mortgage premiums, especially for low-credit-quality borrowers. The right panel shows that borrower net utility is lower with refinancing, and some risky borrowers are excluded from credit under free refinancing.

Economic message. Free refinancing creates an ex ante adverse-selection cost. Penalties reduce rates and expand access.

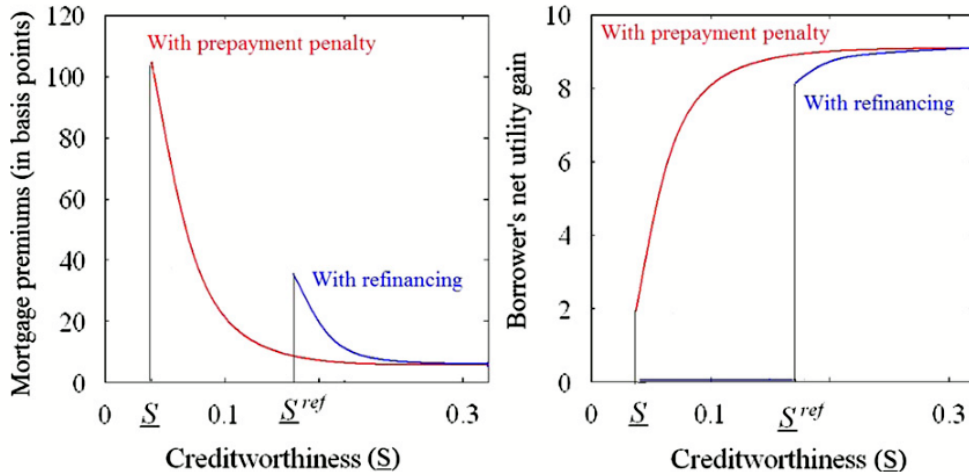


Figure 1: Mortgage premiums and the borrower's net utility gain from homeownership. This figure plots the mortgage premiums in basis points (left

Figure 1: Mortgage premiums and borrower net utility gain from homeownership

6.2 Figure 2 and Figure 3

Read: Figure 2 shows that penalty use is almost universal among the least creditworthy borrowers and falls with creditworthiness. Figure 3 shows that June 2003 occurs near the low point of mortgage rates, supporting the empirical focus on credit-quality rather than rate-driven refinancing.

Economic message. The model predicts penalty concentration among risky borrowers; the empirical timing limits a major alternative refinancing motive.

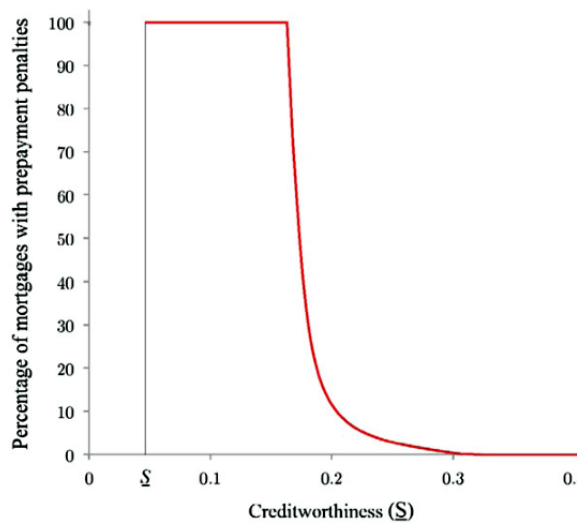
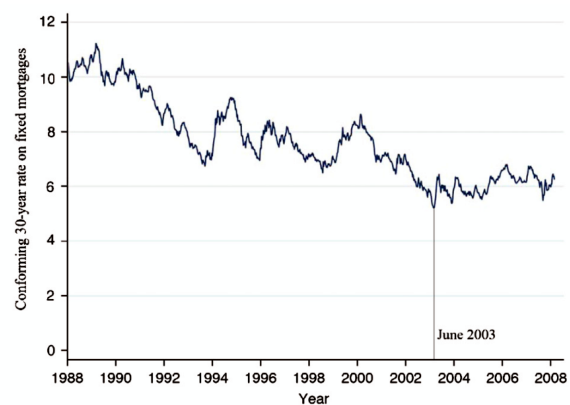


Fig. 2. Model with mobility benefits: percentage of mortgages with prepayment penalties. This figure plots the fraction of loans with

(a) Figure 2: Penalty share by creditworthiness



(b) Figure 3: Conforming 30-year FRM rates

6.3 Table 1: Variables description

Read: The table defines payoff outcomes, HPI growth, prepayment penalty status, coupon rates, credit scores, lien status, CLTV, loan purpose, documentation, and term.

Economic message. It makes the model's theoretical variables empirically measurable in loan-level data.

Variable	Description
Pays Off in X Qs	One if the loan pays off in full in the first X (eight or 16) quarters from origination and zero otherwise
Average Quarterly HPI Growth	The quarterly, real housing price index at the respective horizon calculated using Office of Federal Housing Enterprise Oversight MSA-level data
Prepayment Penalty	One if the loan contains a prepayment penalty and zero otherwise
Coupon Rate	Monthly interest rate charged to the borrower
Origination Amount	Origination amount in thousands of dollars
FICO	Credit score of the borrower using the FICO credit profile
Vantage	Credit score of the borrower using the Vantage credit profile
Percentage Change in Vantage	Captures the difference (in percentage terms) between the last observed Vantage credit score of the borrower before the month of prepayment and initial Vantage at loan origination
Lien	Either <i>Senior</i> for loans in the first position or <i>Subordinate</i> for loans that are not in the senior position
Senior Has Juniors	One when the first lien loan is known to have an associated junior lien and zero otherwise
Loan Is Subprime	One when the loan is classified as subprime and zero otherwise
CLTV	Combined loan-to-value: for junior liens, the loan-to-value (LTV) ratio calculated as the sum of the value of all liens against the home over the value of the home; for senior (first) lien loans, due to the fact that we do not observe the value of additional liens, we use as the CLTV the ratio of the first loan amount to the value of the home
Loan Purpose	Either <i>For Purchase</i> , <i>Cash Out Refinance</i> , or <i>No Cash Out Refinance</i>
Documentation	Variable that is <i>Full Documentation</i> , <i>Low Documentation</i> , or <i>No Documentation</i>
Term	Categorical variable for the length of the term, either 10 years, 15 years, or 30 years

Table 1: Variables description

6.4 Table 2 and Table 4

Read: Table 2 shows that no-penalty subprime loans have much higher payoff rates and coupon rates than prime loans. Table 4 shows that HPI growth strongly predicts subprime prepayment but weakly predicts prime prepayment.

Economic message. Risky borrowers are both more exposed to prepayment incentives and more sensitive to positive credit shocks.

Table 2

Summary of fixed rate mortgage (FRM) loans with no prepayment penalties.

The table covers owner occupied FRM loans without a prepayment penalty originated during June 2003 from the LoanPerformance (LI) database.

Variables	Subprime		Prime	
	Mean	Standard Deviation	Mean	Standard Deviation
Pays Off in 16 Qs	0.702	0.457	0.314	0.464
Average Quarterly HPI Growth	0.011	0.009	0.018	0.009
Coupon Rate	9.075	1.905	5.478	0.361
Origination Amount	110.6	88.8	485.5	164.9
CLTV	80	15	63	15
FICO	574.4	32.045	738.9	40.932
No Cash Refinance	0.124	0.329	0.636	0.481
Cash Out Refinance	0.648	0.478	0.193	0.394
Low Documentation	0.18	0.384	0.289	0.453
No Documentation	0.004	0.063	0.067	0.25
15 Year Term	0.132	0.338	0.395	0.489
20 Year Term	0.162	0.369	0.008	0.09
Senior has Juniors	0.009	0.093	0.09	0.286
Subordinate Lien	0.205	0.404	0	0
Number of Loans	2,517		9,631	

(a) Table 2: No-penalty FRM summary

Table 4

Effect of house price growth for prepayments.

The table covers owner occupied FRMs without a prepayment penalty originated during June 2003 from the LoanPerformance database. Payoff is defined as voluntary prepayment. House price growth is calculated using Office of Federal Housing Enterprise Oversight MSA-level data. Other control include coupon rate, FICO score, origination amount, CLTV (combined loan-to-value), loan purpose, documentation level, term length, and junior or senior status. Coefficients reported are the marginal effects from a logistic regression multiplied by the standard deviation of the right-hand side variable. t statistics are in parentheses.

	Subprime	Prime
	Pays Off in 16 Qs	Pays Off in 16 Qs
Dependent variable mean	0.702	0.314
Average Quarterly HPI Growth	0.083 (7.19)	0.006 (1.18)
Other Controls	Yes	Yes
Number of Loans	2,517	9,628

(b) Table 4: HPI growth and prepayment

6.5 Table 3: Summary statistics for all subprime FRMs

Read: About 72.2% of subprime FRMs carry prepayment penalties. Junior liens are riskier, with higher coupon rates, higher CLTV, and higher payoff rates.

Economic message. The table establishes that penalties are common among subprime borrowers and that junior liens provide a riskier subgroup for testing rate discounts.

Table 3
Summary statistics for fixed rate mortgage (FRM) subprime loans both with and without prepayment penalties.
The table covers owner occupied FRMs originated during June 2003 from the LoanPerformance (LP) database.

Variables	All liens		Senior liens		Junior liens	
	Mean	Standard Deviation	Mean	Standard Deviation	Mean	Standard Deviation
Pays Off in 16 Qs	0.719	0.448	0.697	0.459	0.903	0.296
Average Quarterly HPI Growth in 16 Qs	0.016	0.012	0.016	0.012	0.016	0.012
Prepayment Penalty	0.722	0.448	0.748	0.434	0.53	0.499
Coupon Rate	8.363	1.635	7.956	1.204	11.304	1.281
Origination Amount	133.8	92.44	147.09	90.74	38.42	20.47
CLTV	78	16	75	15	96	10
FICO	576.6	31.218	574.61	31.613	590.7	23.850
No Cash Refinance	0.14	0.346	0.152	0.359	0.047	0.212
Cash Out Refinance	0.693	0.461	0.749	0.434	0.292	0.455
Low Documentation	0.183	0.387	0.2	0.4	0.061	0.239
No Documentation	0.005	0.072	0.005	0.073	0.004	0.06
15 Year Term	0.109	0.312	0.092	0.289	0.233	0.423
20 Year Term	0.12	0.325	0.054	0.226	0.597	0.491
Senior Has Juniors	0.013	0.115	0.015	0.123	0	0
Subordinate Lien	0.122	0.327	0	0	1	0
Number of Loans	9,046		7,945		1,101	

Table 3: Summary statistics for all subprime FRMs

6.6 Table 5 and Figure 4

Read: Table 5 shows that HPI growth and Vantage-score improvements both strongly predict prepayment. Figure 4 shows that penalty use is high among subprime FRMs and very low among prime FRMs from 2003 to 2006.

Economic message. Borrower credit improvement, not just interest rates, drives prepayment; penalty concentration among subprime borrowers is persistent.

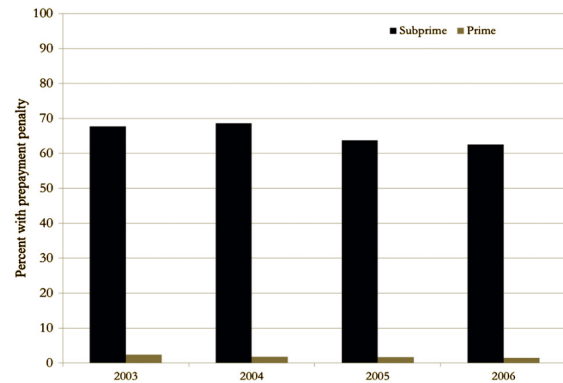
Table 5

Effect of house price growth and the borrowers' credit score changes for prepayments.

The table contains a sample of subprime FRMs with no indication of prepayment penalty originated during 2005 from the BlackBox-Equifax database. Payoff is defined as voluntary prepayment. House price growth is calculated using Office of Federal Housing Enterprise Oversight MSA-level data. The change in Vantage is the variable capturing the difference (in percentage terms) between the last observed Vantage score before the month of prepayment and initial Vantage at loan origination. Other controls include interest rate, original vantage, original balance, CLTV (combined loan-to-value), loan purpose, documentation level, term length and junior or senior status. Coefficients reported are marginal effects from a probit regression. *t*-statistics are in parentheses.

Dependent variable	Pays Off in 8 Qs
Dependent variable mean	0.215
Average Quarterly HPI Growth	0.102 (33.25)
Percentage Change in Vantage	0.00357 (9.59)
Other Controls	Yes
Number of Loans	12,103

(a) Table 5: HPI growth, Vantage changes, and prepayment



(b) Figure 4: Penalty use by origination year

6.7 Table 6: Mortgage coupon and prepayment penalty for subprime loans

Read: Penalty loans have lower rates after controls: -0.195 percentage points for all liens, -0.139 for senior liens, and -0.624 for junior liens.

Economic message. Penalties buy down interest rates, and the discount is largest for the riskiest loan type.

Table 6
Mortgage coupon and prepayment penalty for subprime loans.
The table covers owner occupied fixed rate mortgage loans originated during June 2003 from the LoanPerformance database. Coupon rate is the monthly interest rate charged to the borrower (in percentage terms). Prepayment penalty is a dummy variable that takes value of one if the loan contain a prepayment penalty and is zero otherwise. Other controls include FICO score, origination amount, CLTV (combined loan-to-value), loan purpose, documentation level, term length, junior or senior status, and MSA fixed effects. Coefficients reported are from an ordinary least squares regression. *t*-statistics are in parentheses.

	All liens	Senior liens	Junior liens
Dependent variable	Coupon Rate	Coupon Rate	Coupon Rate
Dependent variable mean	8.363	7.956	11.304
Prepayment Penalty	-0.195 (-6.90)	-0.139 (-4.55)	-0.624 (-7.93)
Other Controls	Yes	Yes	Yes
Number of Loans	9,046	7,045	1,101

Table 6: Mortgage coupon and prepayment penalty for subprime loans

7 Short summary

- Free refinancing creates adverse selection because good-shock borrowers leave the original pool.
- Prepayment penalties restore commitment and support implicit insurance against costly default.
- The benefit is largest for risky borrowers, explaining why penalties are concentrated in subprime mortgages.
- Empirical evidence shows that positive HPI and credit-score shocks predict prepayment, especially for subprime loans.
- Penalty loans carry lower coupon rates, especially for junior liens.

8 Conclusion

8.1 Core conclusion

Prepayment penalties can be welfare-improving in a competitive mortgage market when borrowers are risky and default is costly.

8.2 Economic intuition

Free refinancing allows borrowers with good shocks to leave, weakening risk sharing and forcing higher initial rates. A penalty is a commitment device that can lower rates, reduce default, and expand credit access.

8.3 Incremental contribution

The paper moves the debate from a purely predatory-lending interpretation to a dynamic contracting interpretation. It shows why a restrictive clause can improve welfare ex ante and why the clause should be concentrated among risky borrowers.

8.4 Policy implication

Blanket bans on penalties may remove a useful commitment device. A better approach is stronger disclosure, direct restrictions on unaffordable lending, and contingent penalties that deter refinancing but do not block house sales.

8.5 Limitations

The paper abstracts from behavioral mistakes, monopoly power, and externalities of risky lending. The empirical evidence is consistent with the model but is not randomized assignment of penalty contracts.

8.6 Takeaway

A mortgage term can be restrictive and beneficial at the same time: the ex post restriction can enable ex ante lower rates and access for risky borrowers.